FLATNESS AT POINTS IS UPWARD AND DOWNWARD STABLE ALONG CARTESIAN CHANGE OF BASE, WITH THE FLAT-LOCUS BASE-CHANGE COROLLARY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4467-4492 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 4494-4509, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted A side only (spaces twin incomplete per ruling v3).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Algebra, lemma base-change-flat-up-down; chapter theorem openness-of-flatness for the corollary clause.

Source reference. Original lemma 00MQ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 4509–4509. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0399: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 4446–4449. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

be a cartesian diagram of schemes. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Let $x' \in X'$ with images $x = g'(x')$ and $s' = f'(x')$.

(1) *If $\mathcal{F}$ is flat over S at x, then $(g')^*\mathcal{F}$ is flat over S' at x'.*
(2) *If g is flat at s' and $(g')^*\mathcal{F}$ is flat over S' at x', then $\mathcal{F}$ is flat over S at x.*

In particular, if g is flat, f is locally of finite presentation, and $\mathcal{F}$ is locally of finite presentation, then formation of the open subset of Theorem 0399 commutes with base change.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Consider the commutative diagram of local rings

$$\begin{array}{ccc} \mathcal{O}_{X',x'} & \longleftarrow & \mathcal{O}_{X,x} \\ \uparrow & & \uparrow \\ \mathcal{O}_{S',s'} & \longleftarrow & \mathcal{O}_{S,s} \end{array}$$

Note that $\mathcal{O}_{X',x'}$ is a localization of $\mathcal{O}_{X,x} \otimes_{\mathcal{O}_{S,s}} \mathcal{O}_{S',s'}$, and that $((g')^*\mathcal{F})_{x'}$ is equal to $\mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{O}_{X',x'}$. Hence the lemma follows from Algebra, Lemma 00MQ. $\square$